



AI, IPR, and Economic Development: Nascent Relationships

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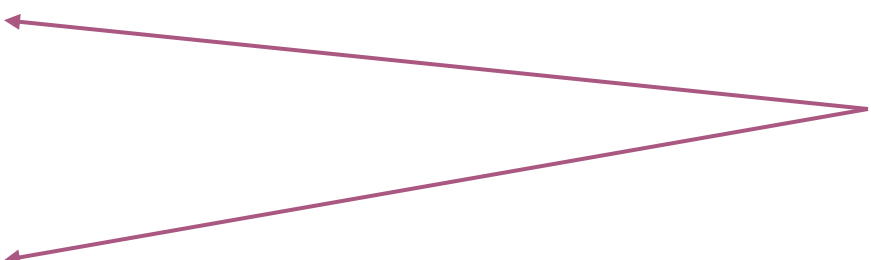
**Sciences Po, Paris Workshop on “Kicking Away the
Innovation Ladder? Intellectual Property Perspectives
on AI Development”, Dec. 11, 2023**

Motivation

- Many benefits (and risks) with AI – economic, social, and others
- Economics literature focuses on effects on productivity, thus far
- Gaps:
 - Little attention to IPRs among economists, considering that IPRs govern the innovation regime behind AI innovations and usage
 - What changes, if any, are needed in IP regimes?
 - How does this affect the Global (North-South) divide?

Conventional View

- Appropriation
 - Let ϵF = sunk cost (innovation)
 - P = price per unit
 - C = cost per unit
 - π = profit per unit
 - Q = quantity sold
- Recouping requires $\pi Q \geq F$

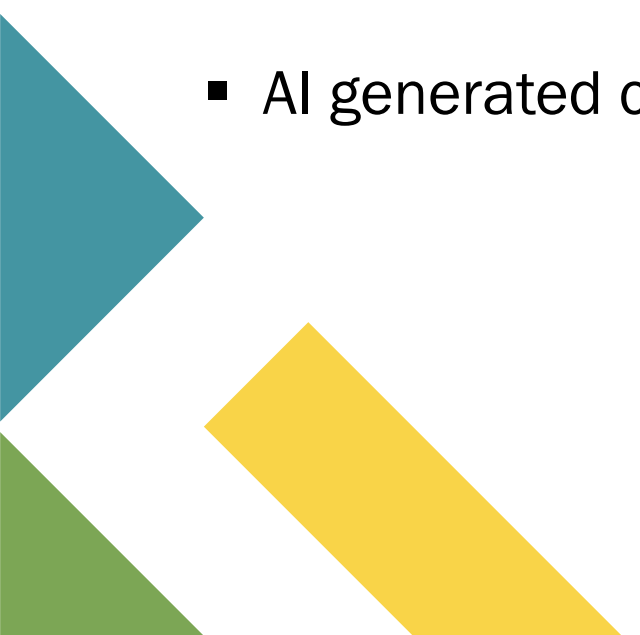


Where IPR often matters

Conventional View

- Optimal IP (non-linear)
- Case for differentiation of IP strength by
 - Level of Economic Development
 - Industry
 - Type of Product
- Role of AI and how it affects the conventional framework:
 - Imitation risks and innovation boost
 - Hardware and software
 - Introduces new **debates** (not necessarily new solutions)

Debates



AI and Copyrights

- Training Datasets
- AI generated creations

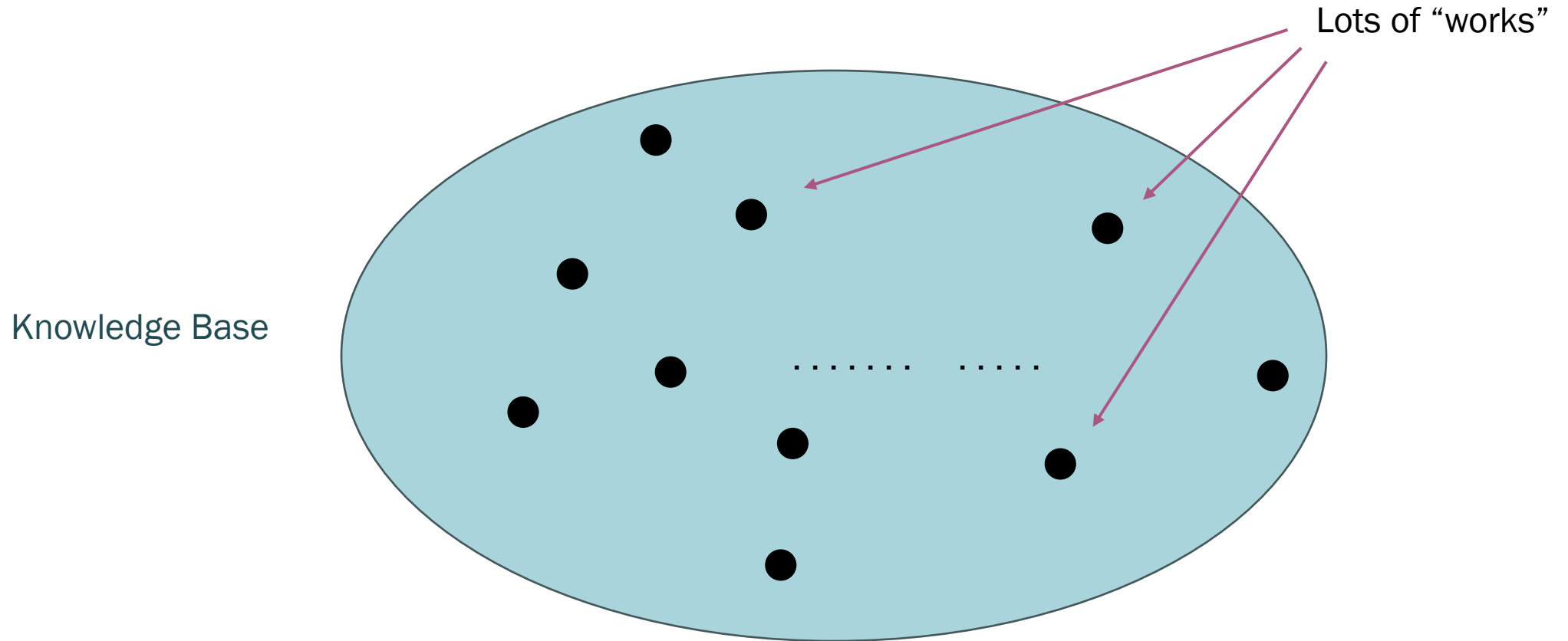
AI and Patents

- Non-obviousness standard
- Patent eligibility & costs of innovation

Common questions

- AI as inventor (output perspective)
- AI training datasets (input perspective)
- Either way, I prefer the answer to be “NO” for both
 - No patent/copyright protection for autonomously or endogenously created innovations or works. Pure economic rent.
 - No additional fees/royalties for works in the training dataset.

Transactions costs



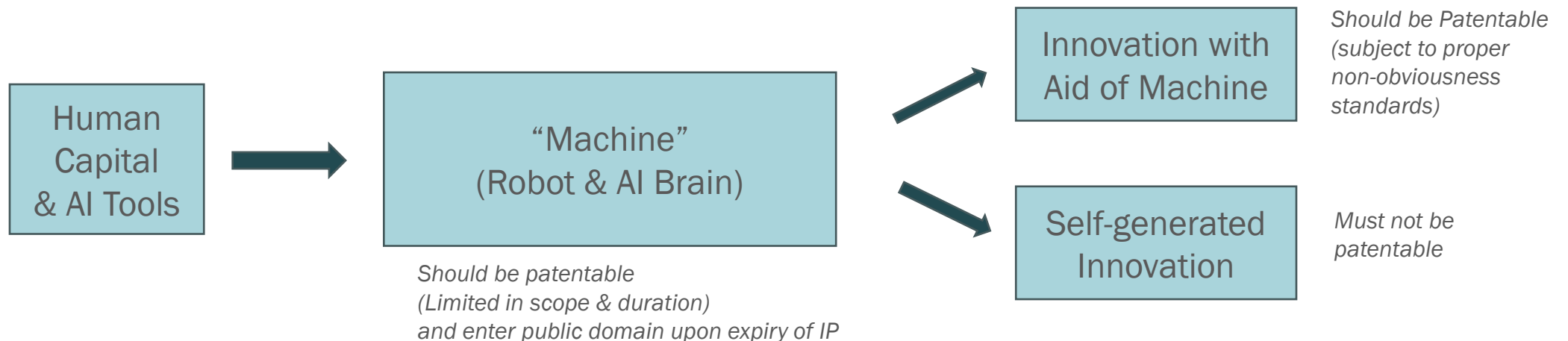
For AI to be effective/accurate, it should be trained on complete knowledge

Case for Public Domain

- Compare to how patent disclosure works
 - In exchange for exclusive rights, owner reveals knowledge
 - That knowledge becomes part of the public domain (the big **circle** in the previous slide)
 - Patent right protects only the product or technology based on the knowledge, not the knowledge itself. *(The patented invention is an excludable good, but the underlying knowledge remains non-rival and non excludable).*
- Where is the copyright infringement?
 - Is the work in the training dataset being exploited commercially? Is it a work of authorship (see C. Craig's writings)? Or simply a source of information? Is the training dataset unlawfully using 'expressions' thereby?

Other issues: i. obviousness & ii. AI as inventor

- AI will likely make new innovations more ‘obvious’
 - Raises bar for patentability
 - Reduces €F. Obviates need for strong patents for purposes of appropriability.
- Autonomous Innovation



- Futurist outlook: possibility of many (similar or exact) independent innovations. Smart AI will search for best & few “bests” exist.

Impact on Economic Development

IP Licensing Receipts of Fees & Royalties (Exports)

Country Group	2005	2010	2015	2020	2022
A. Developed	209.4	273.2	323.6	348.8	346.9
B. Developing	3.1	5.4	17.7	25.4	31.8
Ratio A/B	68	51	18	14	11

- Technological gap will widen, at least temporarily, if not permanently unless AI benefits and tools diffuse globally. AI models, tools, platforms, assets, innovations, firms, research universities, IP rights holders mostly concentrated in N. Think about competition policy and standards setting.
- Skills gap between N & S in AI will likely widen too. Focus on education and labor market policies.
- How to get S participation: value chains, open-source projects, tech transfers, joint ventures, access to knowledge (A2K).
 - What works for the South: reduce trade/investment barriers, keep training datasets as part of public domain (or exempt from copyright assertion). Publicly support generative AI training in open innovation communities. Encourage AI innovations for S needs and provide clarity on IP eligibility and rights.